Karthik Muthumari

Full stack developer - React, NextJs, Nodejs, GraphQL, Apollo Federation, TypegraphQL, AWS (Certified Solution Architect), IaC - CDK, Terraform, SAM CLI, Devops - Gitlabs / Github Actions / Docker / CI CD, Hybrid mobile developer (Ionic 2 & 3, React Native)

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EXPERIENCE

Mindera– Full stack Engineer (5 years)
SEPTEMBER 2021 - PRESENT

Wipro Technologies – Senior Software Engineer (8.5 years)
OCTOBER 2012 - AUGUST 2021

RECENT PROJECTS

Mindera (September 2021 - Present)

*Kingfisher (September 2025 - Present) - **Retail / E-Commerce***

Tax Engine, Kingfisher PLC – Senior Engineer (3 months)

Tax calculation platform supporting basket and checkout services across multiple Kingfisher banners. Backend architecture – **Node.js, TypeScript, REST APIs**. Integrated with enterprise tax data feeds containing **tax plans, rates and jurisdictions**, with event/data ingestion through **Kafka**. Designed service components for tax data ingestion, persistence and runtime tax calculation. Cloud hosted – AWS: **Lambda, CloudWatch**. Observability – **DataDog.**. IaC – **CDK**.

AI-assisted engineering – **Claude Code**. Applied **Spec-Driven Development (SDD)** with structured Markdown specifications, project-level AI guardrails and reusable development instructions. Built a **local RAG-based knowledge retrieval** approach using **vector embeddings, LanceDB, BM25 lexical search and Reciprocal Rank Fusion (RRF)** to provide relevant project context to LLMs during development.

Basket Orchestrator, Kingfisher PLC – Senior Engineer (8 months)

Basket Orchestrator - **Node js, TypeScript, GraphQL, Apollo Federation, Playwright**. Currently migrating legacy REST endpoints. Cloud hosted – AWS: **API Gateway**, **Lambda, Fargate, CloudWatch**. Observability – **DataDog** & **Dynatrace**. IaC – **CDK**. AI-assisted development using **GitHub Copilot, Codex & GitLab Duo** for code generation, refactoring, test creation and developer productivity.

SKILLS

React.js + a11y

Node.js

Next Js

GraphQL + TypeGraphQL

Apollo Federation

AWS - CF, S3, EC2, Lambda, WAF, Elasticache, SQS, API Gateway, Step Function

Jest, TDD

Playwright

Devops - Github Actions, GitLab Pipelines, Jenkins, Docker, CICD

Git + Branching strategy

IaC - CDK, SAM CLI

Google Tag Manager

Ionic 2 & 3

Angular 2 & 4

React Native + Expo

Ansible

TOOLS

Claude code

Co-Pilot

Cursor

SDD - OpenSpec

Selfridges (September 2021 - September 2025) - Fashion E-Commerce / Retail Tech

PLP (Product Listing Page), Selfridges – Frontend developer (1 year)

Product listing UI Application (**MFE**) - **Next.js, TypeScript, GraphQL**. Integrated features – sponsored products, product listings, quotes, component library. Cloud hosted – AWS: **CloudFront, S3, Lambda, CloudWatch (Logs, RUM) & Cloudflare**. Observability – **Dynatrace**. CMS – **Contentstack**. IaC – **CDK**.

Pricing, Selfridges – Full stack developer (2 years)

Event-driven pricing application. BE architecture - **Nodejs, Typescript**. Event Orchestration - **S3, Lambda, Step Functions, SNS, SQS, DynamoDB**. API Layer - **API gateway, Lambda, Authorisors**. Internal UI for Vendor pricing upload - **Cloudfront, S3, Lambda**. CI/CD - **Github Actions**. IaC - **CDK, Terraform**

VIP, Selfridges – Full stack developer (1 year)

VIP tier Selfridges application. Serverless Web application - **Next.js**, **a11y**. Serverless Aggregator - **Node.js (micro-services)**. Cloud hosted - **AWS - CloudFront, API Gateway, Lambda, SQS, WAF, S3, ALB, Secret Manager. CI CD - Github Actions.** Observability – **New Relic.** IaC **- CDK**

Personal shopper, Selfridges – Frontend developer (1 year)

Personal shopper application. Web application - **Gatsby.js**, **a11y**. CMS - **Strapi**. **AWS - CloudFront, WAF. CI/CD - Github Actions,** IaC **- CDK**

Wipro (October 2012 - August 2021)

Capital One - FinTech / Banking Technology (ONSITE UK, March 2018 - Aug 2021)

Quick Check, CapitalOne – Full stack developer (ONSITE UK, Sept 2019 - Aug 2021)

Credit eligibility checker. Web application - **React.js**, **a11y**. Orchestration - **Node.js (micro-services)**. Cloud hosted - **AWS - CloudFront, WAF, S3, ALB, Lambda with Hot site resiliency (operated in two regions). CI CD - Jenkins, Docker.**

CreditWise, CapitalOne – Full stack developer (ONSITE UK, March 2018 - Sept 2019)

Credit score tool. Web application - **React.js, a11y**. Orchestration - **Node.js**. Cloud hosted - **AWS - CloudFront, WAF, S3, ALB, EC2 instance, Elasticache with standby resiliency (active - passive regions). CI CD - Jenkins, Docker.**

US LBM - Retail / Supply Chain / E-Commerce

Hines Supply, US LBM – Hybrid mobile - Web app developer (India, Jan 2016 - Feb 2018)

Payment, delivery, invoice Tracking application. Framework: **Ionic framework, Angular.js**. Devices: **Android mobiles, Tablets, iPones, iPad** and across **all the browsers** in desktop. Features: **Push notification, Camera, Navigation, File & Files transfer.**

AWARDS

Certified Solution Architect
AWS Certified Solution Architect Associate - 2021

IBM Hackathon - WINNER
Developed Mobile applications integrated with chat bot - IBM Watson & Bluemix - 2017

EDUCATION

B.E. | 2012 | Anna university, Chennai, India
MAJOR: ECE
JEPPIAAR ENGINEERING COLLEGE

LANGUAGES

English, Tamil

COACH - Fashion E-Commerce / Retail Tech

Visual Merchandising, COACH – Hybrid mobile app developer (India, Aug 2015 - Jan 2016)

Merchandising application. Framework: **Ionic 3, Angular 4, SASS.** Device: **Android mobiles, Tablets, iPhones, iPad.** Features: **Barcode, Offline catalog storage, Camera, Photo upload, New feeds.**

Cardinal Health - Healthcare / MedTech / Supply Chain

Pack manager, Cardinal Health – Hybrid mobile app developer (India, Jan 2015 - Aug 2015)

Health care business application. Framework: **Angular UI, Angular.js**. Device: **iPads & Windows Surface Pro.** Features: **Lazy loading, QR/Barcode, E-signature**